# MAMMO LENS Review 1 Proposal

## Mammogram classification with global and local evidence

CAI3400 Image Processing and Analysis
Presidency University Bengaluru
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This proposal presents the research question, primary literature, audited dataset and preliminary real-data results. The next phase evaluates deep global/local models and probability reliability across density categories and controlled degradation.

# Project proposal and objectives

MAMMO-LENS investigates retrospective benign versus malignant classification of selected abnormal mammograms. The proposed experiment asks whether global context and local patches improve performance, and whether any benefit remains reliable across recorded density categories and controlled image degradation.

The project combines a reproducible local image-analysis pipeline with a browser evidence viewer. Its objectives are to compare global, patch-only and combined features on identical patient partitions; evaluate probability quality and selective prediction; and examine density subgroup and degradation failures. Future cancer-risk prediction requires longitudinal outcomes, which this experiment does not contain.

## Research hypothesis

Deep global/local fusion may improve classification over a global-only reference. Density conditioning may improve reliability in some groups. Both are hypotheses to test, rather than established advantages. Existing GMIC work already combines global and local information [6], so the proposed contribution is the controlled reliability experiment and transparent cohort audit.

## Scope

The primary endpoint is image-level pathology classification. An exploratory patient endpoint takes the maximum image score and labels a patient positive if any included image is malignant. This selected abnormal-case cohort cannot establish general screening prevalence, population predictive value or clinical readiness.

# Literature review

## [1] Lee et al. (2017)

A curated mammography data set for use in computer-aided detection and diagnosis research

Curated DDSM cases with pathology, mass/calcification annotations and breast-density descriptors. Defines the dataset and label provenance. Our audited Kaggle JPEG subset differs from the original DICOM collection. Selected abnormal cases support retrospective classification. Future outcomes and screening prevalence require other data.

## [2] He et al. (2016)

Deep Residual Learning for Image Recognition

Residual connections ease optimization of deep image networks. The paper evaluates natural-image recognition. Provides the ResNet family used for our planned ImageNet-initialized ResNet18 encoder. Natural-image benchmark success does not establish mammography performance.

## [3] Ilse et al. (2018)

Attention-based Deep Multiple Instance Learning

Learns a permutation-invariant attention operator for bags of instances, with experiments including histopathology. Provides gated attention for pooling selected patch embeddings. Different tissue/task and data. Attention weights do not establish causal or clinical explanations.

## [4] Guo et al. (2017)

On Calibration of Modern Neural Networks

Studies probability calibration and evaluates temperature scaling on image and document classification. Motivates fitting one temperature using validation predictions and evaluating probability quality separately. Calibration must be re-evaluated under dataset or image-quality shift.

## [5] Li Shen et al. (2019)

Deep Learning to Improve Breast Cancer Early Detection on Screening Mammography

Uses annotated patches to initialize an all-convolutional whole-image classifier, with DDSM development and INbreast transfer. Establishes prior patch-to-image mammography learning and motivates a controlled local/global comparison. Its annotations, cohort and transfer protocol differ from our experiment. Headline scores are not comparable.

## [6] Yiqiu Shen et al. (2021)

An interpretable classifier for high-resolution breast cancer screening images utilizing weakly supervised localization

GMIC combines a lightweight global network, saliency-selected high-resolution patches, gated attention and fusion on NYU screening mammograms. Closest architectural precedent. It motivates retaining local detail and explicitly crediting existing fusion and attention techniques. Our current patches come from a 256-square image and use texture selection. We cannot claim equivalent localization or clinical validation.

The reviewed sources establish the component methods and close mammography precedents. They do not establish that this project's exact combination is novel. Additional recent and density-specific literature remains part of the continuing review. Cross-paper AUROC comparisons are inappropriate without matched cohorts, supervision and split protocols.

# Dataset audit and experimental protocol

Source: awsaf49/cbis-ddsm-breast-cancer-image-dataset, a Kaggle JPEG export of CBIS-DDSM. Original dataset citation: Sawyer-Lee et al. (2016), TCIA, DOI 10.7937/K9/TCIA.2016.7O02S9CY. The downloaded archive is 5,318,997,088 bytes. Local SHA-256 receipt: 3db02da638d4848068ef9940b7631a6307 efaa9829c78681b7d2c0bd87c31dd0. This hash records our download and is not a publisher-signed checksum.

| Audit field | Result |
|---|---|
| Full images / patients | 2,955 / 1,554 |
| Benign / malignant images | 1,653 / 1,302 |
| Density known / unknown | 2,953 / 2 |
| Missing mapped JPEGs | 0 |
| Training views excluded for test-patient overlap | 64 |
| Duplicate groups / images excluded | 42 / 84 |

The adapter maps each case-description series UID to dicom_info.csv. Blank series descriptions require exact UID mapping. Both BENIGN and BENIGN_WITHOUT_CALLBACK map to label 0. MALIGNANT maps to 1. Repeated lesion descriptions for one image aggregate with an any-malignant rule. All members of exact duplicate-byte groups are excluded. This detects identical files, but cannot exclude every recompressed or near-duplicate image.

| Partition | Images | Patients |
|---|---|---|
| Train | 1,740 | 905 |
| Validation | 578 | 302 |
| Test | 637 | 347 |

The test cohort retains official test images after exclusions. Validation is a fixed patient-group split of retained training patients using seed 42. Every patient belongs to one partition. Training patients number 905, validation 302 and test 347. The declared exclusions mean this is a derived cohort, not an unchanged official benchmark.

# Methodology and evaluation

Each image is converted to grayscale while preserving intensity range, normalized between its first and 99.5th intensity percentiles and resized to 256 by 256 pixels. Four 64 by 64 grid patches with high intensity variation are selected after background filtering. Texture selection does not locate lesions.

The completed handcrafted models use intensity histograms, intensity statistics, quantiles and gradient magnitudes. Global mode uses the whole image. Patch mode pools patch feature means and maxima. Combined mode concatenates both. A StandardScaler and logistic regression fit only the training features.

The implemented deep model uses a shared ImageNet-initialized ResNet18 [2]. It encodes a 224-square global view and four 128-square resized patches. Gated attention [3] pools patch embeddings. The classification head consumes global, local or concatenated features. A supervised density head uses known density labels. Optional density-conditioned fusion consumes predicted density probabilities during inference. Real deep checkpoints and scores remain pending.

For the planned initial deep comparison, use a frozen encoder, batch size 8, seed 42 and 10 epochs, holding budget and selection rule constant. Validation log loss selects the checkpoint. Validation-only temperature scaling [4] calibrates scores. Deep tuning, higher-resolution alternatives and runtime optimization remain future work.

Report AUROC, sensitivity and specificity at a stated threshold, Brier score, density subgroup counts and coverage versus error. Current intervals resample patients with all their views in 200 cluster bootstrap replicates. Evaluate blur radius 2 and Gaussian noise standard deviation 0.05 as paired perturbations at working resolution. Patch masking replaces a selected region with median intensity and recomputes the pipeline, so score changes cannot isolate causal importance.

# Preliminary measured results

| Feature mode | AUROC | 95% cluster CI | Brier | Sensitivity | Specificity |
|---|---|---|---|---|---|
| global | 0.6249 | 0.5749 to 0.6785 | 0.2359 | 0.4457 | 0.7282 |
| patch | 0.5907 | 0.5387 to 0.6408 | 0.2385 | 0.4225 | 0.7045 |
| combined | 0.6198 | 0.5777 to 0.6657 | 0.2379 | 0.4651 | 0.7018 |

All results above come from real CBIS-DDSM baseline artifacts on 637 held-out images from 347 patients. Sensitivity and specificity use threshold 0.5. The intervals describe image AUROC uncertainty using patient-cluster resampling. These are preliminary results from one split, not clinical validation.

The global reference has the strongest point estimate. Combined features have not demonstrated an improvement, and overlapping intervals alone do not constitute a paired significance test. Patch-only

features perform worse by point estimate. We will preserve this negative finding rather than assume fusion must help.

### Reliability observations

For the global baseline, recorded density category 4 has AUROC 0.4895 on 113 test images, compared with 0.6964 for category 2 on 258 images. Small groups and cohort composition limit interpretation. The result does not demonstrate a causal density effect.

Global AUROC under blur is 0.6197. Added noise reduces AUROC to 0.5902 and raises Brier score from 0.2359 to 0.3969. At threshold 0.5, noise produces sensitivity 0 and specificity 1 because all outputs fall on the benign side. This highlights a severe perturbation failure.

For global predictions, confidence 0.6 retains 202 of 637 images with error 0.3020, versus error 0.3862 at full coverage. Confidence 0.8 retains only 8 images with error 0.75. Therefore increased confidence is not a dependable abstention rule here. Operational thresholds must be chosen on validation data.

The baseline test results have already been examined. Future changes motivated by these observations must be described as exploratory. Strong claims about generalization need a fresh external cohort or a prospectively specified evaluation.

## Implementation status and work schedule

Completed local assets include the FastAPI backend, browser interface, image processing, model registry, patient-safe manifest and three trained handcrafted models. The last development session passed 15 automated tests. The local machine has an Intel Core i5-13420H, about 16 GB RAM and an RTX 4050 Laptop GPU with 6 GB VRAM. A separate CUDA environment has been configured. Hardware observations and test results refer to the 26 October session.

Deep model code exists, but the interrupted real deep run saved no checkpoint. Repeated decoding of large JPEGs limited runtime. The next engineering tasks are deterministic preprocessing cache validation, checkpoint resume and measured epoch timing before the full comparison.

| Milestone | Planned evidence |
|---|---|
| 5 November 2026 Review 1 | Proposal, literature, audited cohort and preliminary baselines |
| 17 November 2026 Review 2 | Deep ablations, runtime measurements and intermediate findings |
| 27 November 2026 Final | Complete evaluation, error analysis, final report and demo |

The research plan records Review 1 as 20%, Review 2 as 30% and the final review as 50%. Faculty/LMS instructions should confirm dates, submission format and any title-page requirements. The 15 to 20 page requirement in the saved plan applies to the final report.

## Limitations and responsible interpretation

Major limitations are aggressive resizing, heuristic patch selection, one retrospective dataset, restricted abnormal-case sampling and incomplete external validation. Scanner robustness, clinical diagnosis and longitudinal risk remain outside demonstrated capabilities. All interfaces and presentations should retain research-only labeling.

## AI assistance acknowledgement

Generative AI assisted implementation, debugging and preparation of review materials. The presenting student is responsible for verifying references, understanding the methodology and explaining results. Acknowledge assistance in the form required by the institution.

# References and reproducibility

[1] Lee et al. (2017). A curated mammography data set for use in computer-aided detection and diagnosis research. https://doi.org/10.1038/sdata.2017.177

[2] He et al. (2016). Deep Residual Learning for Image Recognition. https://arxiv.org/abs/1512.03385

[3] Ilse et al. (2018). Attention-based Deep Multiple Instance Learning. https://proceedings.mlr.press/v80/ilse18a.html

[4] Guo et al. (2017). On Calibration of Modern Neural Networks. https://proceedings.mlr.press/v70/guo17a.html

[5] Li Shen et al. (2019). Deep Learning to Improve Breast Cancer Early Detection on Screening Mammography. https://arxiv.org/abs/1708.09427

[6] Yiqiu Shen et al. (2021). An interpretable classifier for high-resolution breast cancer screening images utilizing weakly supervised localization. https://arxiv.org/abs/2002.07613

Dataset: TCIA CBIS-DDSM. Export: Kaggle CBIS-DDSM JPEG export.

Local evidence: data/manifest.audit.json, data/cbis-ddsm/download-receipt.json and artifacts/real-baseline-{global,patch,combined}/metrics.json. Code milestone: 345a6be. Raw data, patient identifiers and model binaries remain outside Git.

Each baseline uses the same data/manifest.csv, mode global, patch or combined, seed 42 and a distinct output directory. The CLI entry point is mammo_lens.cli. Verify partition identity through each run's split_manifest.json. Source links checked on 4 November 2026. This literature summary covers selected primary work and is not exhaustive.